

1.3 State the order of operations for mathematics calculations

Consider the expression $3 + 4 \cdot 2$

- a) If the addition is performed first, the answer is _____.
- b) If the multiplication is performed first, the answer is _____.

Both answers cannot, of course, be correct.

In this problem, the correct answer is _____.

When dealing with arithmetic or algebraic expressions, the following convention is followed.

- Brackets are evaluated first. If more than one set is present, we evaluate from the inside out.
- Exponents
- Multiplication and Division (from left to right)
- Addition and Subtraction (from left to right)

Note

This sequence of calculations is often referred to as BEDMAS.

Evaluate each expression below. Show all calculations.

a) $12 \times 5 - 2$

b) $12 - 5 \times 2$

c) $10 + 16 \div 2^3$

d) $10 + (8 \div 2)^4$

$$e) 6 - 2(3 + 2^2)$$

$$f) 8 + 2(3 + 2)^2$$

$$g) 4[5 - 2(7 - 3)] + 1$$

$$h) 3 + 2(5 - 3^2)^2 - 2$$

$$i) \frac{4 \times 72 + 3^3}{(200 - 5^3) + 5 \times 6}$$